# NATIONAL MATHEMATICS CONTEST
## O-LEVEL PAPER 2.

SATURDAY 23RD JUNE, 2018 9:00 AM - 12:15 PM

**INSTRUCTIONS TO CANDIDATES AND SUPERVISORS**:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are two sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

(c) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(d) ***Indicate your names, class, school and district on all your answer sheets****.*

(e) *Answer scripts of only qualified students should be returned to the contest coordinator, latest Monday 25th June, 2018.*

(f) *ALL participants who qualified for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 28*th July,* 2018 *at 8:00am at Makerere University, Kampala.*

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## SECTION A : 5 marks each

Qn. 1. Let

$$x \star y = 2x^2 + 2xy.$$

If $a$ and $b$ are positive integers such that $a \star b = 9$ and $b \star a = 72$, what is the value of $\frac{a}{b}$. Express your answer as a common fraction.

Qn. 2. Tom can paint a fence in 15 hours by himself. Ben can paint the same fence in 10 hours by himself. Tom started painting the fence alone, but then tricked Ben into painting while he left. In total it took 11 hours to paint the fence. How many hours did Tom work?

Qn. 3. The ratio of Grace's weight to Henry's weight is $2:3$. If the ratio of Henry's weight to Sam's weight is $4:5$, what is the ratio of Grace's to Sam's?

Qn. 4. What is the value of

$$\frac{1}{8} + \log_2 2\sqrt{2\sqrt{2\sqrt{2}}} \ ?$$

Qn. 5. What are the roots of the following equation?

$$\log_2(x+1) + \log_2(x) = 1.$$

Qn. 6. Suppose that

$$3^{2x+1} = 12.$$

Find the value of $\left(\frac{1}{81}\right)^x$.

Qn. 7. What is the reminder when $3^{2018}$ is divided by 7?

Qn. 8. An advertisement reads, "Take 11% off any item which is already discounted 31%." What is the percentage value of the combined discounts?

Qn. 9. A point $P$ is inside the square $ABCD$ with $AB = 8$. The distances between $P$ and the points $A, B$ and the side $CD$ are all $x$. Find the value of $x$.

Qn. 10. Sarah and Jenny are running towards each other. They are 100 meters apart initially. Sarah runs 2 meter/second faster than Jenny, and they meet in 20 seconds. How fast was Jenny running?

## SECTION B : 10 marks each

Qn. 11. A school has two empty rectangular swimming pools, one for adults and the other for children. The volume of the children's pool is 65% of the adult's pool. The staff started supplying water to both pools with the same flow rate. If 80% of the children's pool was filled after $T$ hours, how much more time do we need in terms of a percent of $T$ until 95% of the adult's pool is filled?

Qn. 12. A Laboratory attendent has 100 c.c of a solution that contains 20% acid, the rest being water. Suppose he accidentally adds pure acid to make the solution $33\frac{1}{3}$% acid. How many c.c's of water must he then add to return the solution to 20% acid?

Qn. 13. The Uganda Mathematics contest team for a school had a smooth trip to the contest venue in the morning, but unfortunately repairs on the route they used meant that some section of the route was closed and a diversion created. This made the journey back to school take twice as long. If the new route to school (including the diversion) was 50% longer than the original trip, and the average speed returning was $10km/h$ slower, what was the average speed the team used while going to the contest?

Qn. 14. A $3 \times 3$ magic square contains every positive integer from 1 through 9, with one integer per cell, such that the sums of the numbers in each row, each column and each diagonal are the same. When the ordinary magic square shown is completed, what is the sum of all the possible values of $y$?

| | | |
|---|---|---|
| | $y$ | |
| | 5 | |
| | | $y+1$ |

Qn. 15. In triangle $ABC$, $< BAC = 30^\circ$ and $< ABC$ is a right angle. The point $D$ lies on $AB$ so that $AD = 2cm$ and $< CDB = 45^\circ$. What is the length of $BC$?

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## SECTION A : 5 marks each

Qn. 1. Let

$$x \star y = 2x^2 + 2xy.$$

If $a$ and $b$ are positive integers such that $a \star b = 9$ and $b \star a = 72$, what is the value of $\frac{a}{b}$. Express your answer as a common fraction.

Solution

Since $a \star b = 2a^2 + 2ab = 9$ and $b \star a = 2b^2 + 2ab = 72$ then $\frac{2a(a+b)}{2b(b+a)} = \frac{9}{72} \Rightarrow \frac{a}{b} = \frac{1}{8}$ since $a + b > 0$.

Qn. 2. Tom can paint a fence in 15 hours by himself. Ben can paint the same fence in 10 hours by himself. Tom started painting the fence alone, but then tricked Ben into painting while he left. In total it took 11 hours to paint the fence. How many hours did Tom work?

Solution

Let $x$ be the time Tom worked. Suppose Tom works at a rate $t$ units per hour and Ben works at a rate of $b$ units per hour. Let the total work required to paint the fence be $P$. Then $15t = p$ and $10b = p$. If the total time taken to paint the fence by both was 11 hours, then $x \cdot t + (11-x) \cdot b = p$ and therefore $x \cdot \frac{p}{15} + (11-x) \cdot \frac{p}{10} = p$. Whence $x \cdot \frac{1}{15} + (11-x) \cdot \frac{1}{10} = 1 \Rightarrow x = 3$.

Qn. 3. The ratio of Grace's weight to Henry's weight is 2 : 3. If the ratio of Henry's weight to Sam's weight is 4 : 5, what is the ratio of Grace's to Sam's?

Solution

Let the weight of Grace be $G$, of Henry be $H$ and of Sam be $S$ then $G : H = 2 : 3$ and $H : S = 4 : 5$. Let the total weight of Grace and Henry be $W_1$ and the total weight of Henry and Sam be $W_2$. Then $G + H = W_1$ and $S + H = W_2$. From the ratios it then implies that $\frac{3}{5}W_1 + \frac{5}{9}W_2 = W_2$ thus $W_1 = \frac{20}{27}W_2$. The required ratio is $\frac{G}{S} = \frac{\frac{2}{5}W_1}{\frac{5}{9}W_2} = \frac{\frac{2}{5}\cdot\frac{20}{27}\cdot W_2}{\frac{5}{9}W_2} = \frac{8}{15}$.

Qn. 4. What is the value of

$$\frac{1}{8} + \log_2 2\sqrt{2\sqrt{2\sqrt{2}}} \ ?$$

Solution

$\sqrt{2} = 2^{\frac{1}{2}} \Rightarrow 2\sqrt{2} = 2^{\frac{3}{2}}$, in the same way $2\sqrt{2\sqrt{2\sqrt{2}}} = 2^{\frac{15}{8}}$ so $\log_2 2\sqrt{2\sqrt{2\sqrt{2}}} = \log_2 2^{\frac{15}{8}} = \frac{15}{8}$ thus $\frac{1}{8} + \log_2 2\sqrt{2\sqrt{2\sqrt{2}}}$

Qn. 5. What are the roots of the following equation?

$$\log_2(x+1) + \log_2(x) = 1.$$

Solution

$\log_2(x+1) + \log_2(x) = 1 \Leftrightarrow \log_2\left(\frac{x(x+1)}{2}\right) = 0 \Rightarrow x^2 + x - 2 = 0 \Leftrightarrow x = -2, 1$. The only valid solution is $x = 1$

Qn. 6. Suppose that

$$3^{2x+1} = 12.$$

Find the value of $\left(\frac{1}{81}\right)^x$.

Solution

Clearly $3^{2x+1} = 12 \Leftrightarrow 3^{2x} = 4$. Since $81 = 3^4$ then $\left(\frac{1}{81}\right)^x = \left(\frac{1}{3^4}\right)^x \Rightarrow \left(\frac{1}{81}\right)^x = 3^{-4x} = 4^{-2} = \frac{1}{16}$.

Qn. 7. What is the reminder when $3^{2018}$ is divided by 7?

Solution

Since $3^2 = 2(mod\ 7), 3^3 = 6(mod\ 7), 3^4 = 4(mod\ 7), 3^5 = 5(mod\ 7)$ and $3^6 = 1(mod\ 7)$ and $2018 = 6 \times 336 + 2$ then $3^{2018} = 3^{6\times 336} \times 3^2(mod\ 7) = 3^2(mod\ 7) = 2$.

Qn. 8. An advertisement reads, "Take 11% off any item which is already discounted 31%." What is the percentage value of the combined discounts?

Solution

The price after discount is $\frac{69}{100}p$ where $p$ is the initial price. Price after advertisement is $\frac{89}{100}$ of $\frac{69}{100}p$ thus $0.6141p$ thus the combined discount is $(1 - 0.614) \times 100\% = 38.5\%$.

Qn. 9. A point $P$ is inside the square $ABCD$ with $AB = 8$. The distances between $P$ and the points $A, B$ and the side $CD$ are all $x$. Find the value of $x$.

Solution

Let $M$ be the center of the side $AB$. Now, $AM = 4, PM = 8 - x$ and triangle $AMB$ is a right triangle then $16 = (8 - x)^2 = x^2$ solving for $x$ yields $x = 5$.

Qn. 10. Sarah and Jenny are running towards each other. They are 100 meters apart initially. Sarah runs 2 meter/second faster than Jenny, and they meet in 20 seconds. How fast was Jenny running?

Solution

Let the speed of Jenny be $Sm/s$ then that of Sarah will be $(S + 2)m/s$. Now for sarah, Distance covered $D = (S + 2) \times 20$ and for Jenny $100 - D = 20 \times S$ from where $S = 1.5m/s$.

## SECTION B : 10 marks each

Qn. 11. A school has two empty rectangular swimming pools, one for adults and the other for children. The volume of the children's pool is 65% of the adult's pool. The staff started supplying water to both pools with the same flow rate. If 80% of the children's pool was filled after $T$ hours, how much more time do we need in terms of a percent of $T$ until 95% of the adult's pool is filled?

Solution

When 80% of the children's pool is filled then $80\% \times 65\%$ of the adults pool will be filled that is 52% in $T$ hours. To fill the adult pool to 95% will take $T \times \left(\frac{95}{52}\right) = 1.8269T$ thus it will require an additional 82.69% of $T$.

Qn. 12. A Laboratory attendent has 100 c.c of a solution that contains 20% acid, the rest being water. Suppose he accidentally adds pure acid to make the solution $33\frac{1}{3}\%$ acid. How many c.c's of water must he then add to return the solution to 20% acid?

Solution

Let $x$ be the amount of acid added accidentally then $\dfrac{20 + x}{100 + x} = \dfrac{1}{3} \Rightarrow x = 20cc$. Now, we have $40cc$ of acid out of 120 cc of solution. Let $y$ be the number of $cc$ of water required to add to get back back a percentage of 20%. Thus $\dfrac{y + 80}{120 + y} = \dfrac{8}{10} \Rightarrow y = 80cc$.

Qn. 13. The Uganda Mathematics contest team for a school had a smooth trip to the contest venue in the morning, but unfortunately repairs on the route they used meant that some section of the route was closed and a diversion created. This made the journey back to school take twice as long. If the new route to school (including the diversion) was 50% longer than the original trip, and the average speed returning was $10km/h$ slower, what was the average speed the team used while going to the contest?

Solution

Let $d$ be the distance to the contest and $t$ be time to travel to the contest. Then the average speed going to the contest is $\frac{d}{t}$, which can be denoted as $v$. Returning home involves a distance 50% greater than going that is $(1 + 0.5d) = 1.5d$ and a time twice as long as that of going $(2t)$. The average return speed $= \dfrac{1.5d}{2t} = \dfrac{3}{4}v$ whose value is $10km/hr$ less than $v$. Therefore $\dfrac{3}{4}v = v - 10 \Rightarrow v = 40km/hr$.

Qn. 14. A $3 \times 3$ magic square contains every positive integer from 1 through 9, with one integer per cell, such that the sums of the numbers in each row, each column and each diagonal are the same. When the ordinary magic square shown is completed, what is the sum of all the possible values of $y$?

| | | |
|---|---|---|
| | $y$ | |
| | 5 | |
| | | $y+1$ |

Solution

The sum of the elements in each row, column and diagonal of a $3 \times 3$ magic square must be $\dfrac{1+2+3+\ldots+9}{3} = 15$ on average. For the center column to sum to 15 the bottom must be $10 - y$. Reasoning in a similar way the magic square is filled as below,

| | | |
|---|---|---|
| $9-y$ | $y$ | 6 |
| y + 2 | 5 | $8-y$ |
| 4 | $10-y$ | $y+1$ |

Given that $4, 5, 6$ already appear in the table, then $y \neq 4, 5, 6, y \neq 4, 5, 6$ and $y + 2 = 4, 5, 6$. These three equations imply $y \neq 2, 3, 4, 5, 6$. Also $8 - y, 9 - y$ appear in the table and we can only use numbers from 1 to 9 so $y$ can not be 8 0r 9 hence $y \neq 2, 3, 4, 5, 6, 8, 9$ which leaves only $y = 1, 7$ whose sum is 8.

Qn. 15. In triangle $ABC$, $< BAC = 30°$ and $< ABC$ is a right angle. The point $D$ lies on $AB$ so that $AD = 2cm$ and $< CDB = 45°$. What is the length of $BC$?

Solution

From $\triangle ABC, \tan 30 = \dfrac{x}{x+2} \Rightarrow x = \dfrac{2}{\sqrt{3}-1}$.